

Day - 05

Deep Dive: Measurement & State Collapse

Today, we delve into one of the most counter-intuitive yet fundamental aspects of quantum mechanics: **Measurement**. In the classical world, measuring an object (like checking a speedometer) doesn't change its state. In the quantum world, measurement is an active, irreversible process that forces a system to "choose" a definitive reality.

1. The Mechanics of State Collapse

Before measurement, a qubit can exist in a superposition, mathematically described as a state vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. The values α and β are probability amplitudes.

The Born Rule

The probability of measuring the state $|0\rangle$ is given by the magnitude squared of its amplitude:

$$P(0) = |\alpha|^2$$

Similarly, the probability of measuring $|1\rangle$ is:

$$P(1) = |\beta|^2$$

Because the total probability must equal 1, we always have: $|\alpha|^2 + |\beta|^2 = 1$.

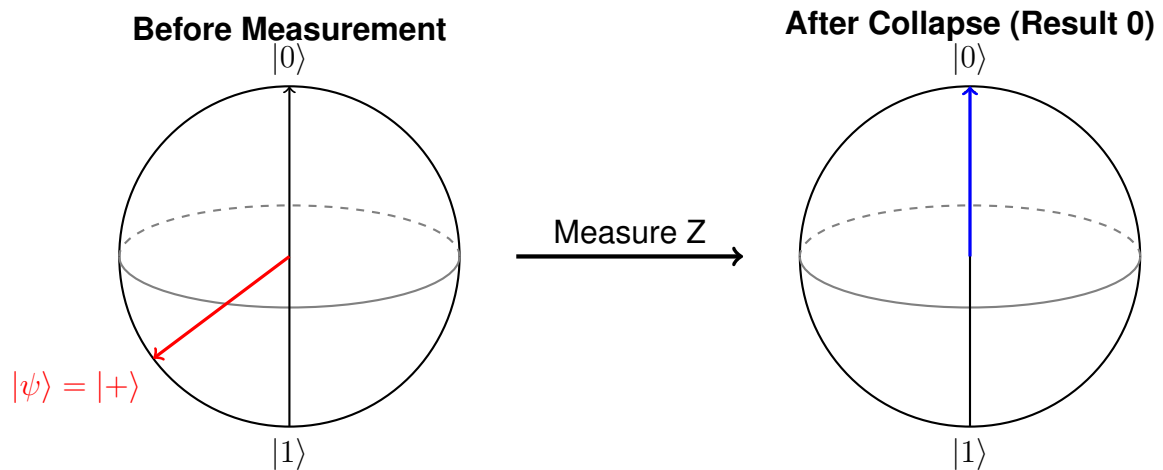
What happens upon measurement? When you measure the qubit, it **collapses** into either $|0\rangle$ or $|1\rangle$ based on those probabilities.

- If the outcome is 0, the new state of the qubit is absolutely $|0\rangle$. The β component is completely destroyed.
- If you immediately measure the same qubit again, the result will be 0 with 100% certainty ($P(0) = |1|^2 = 1$).

This irreversibility is why measurement is almost always placed at the very **end** of a quantum circuit.

2. Visualizing Collapse on the Bloch Sphere

Imagine the Bloch Sphere. A qubit in superposition can point anywhere on the surface (e.g., pointing along the $+X$ axis for the $|+\rangle$ state).



When you measure in the standard (Z) basis, you are asking the qubit: "Are you pointing UP ($|0\rangle$) or DOWN ($|1\rangle$)?" The state vector violently snaps to either the North or South pole, destroying the X -axis information.

3. The Importance of "Shots" (Repeated Execution)

Because quantum measurement is inherently probabilistic, running a quantum circuit exactly one time gives very little information. A single result of '0' doesn't tell you if the state was $|0\rangle$, $|+\rangle$, or $0.99|0\rangle + 0.14|1\rangle$.

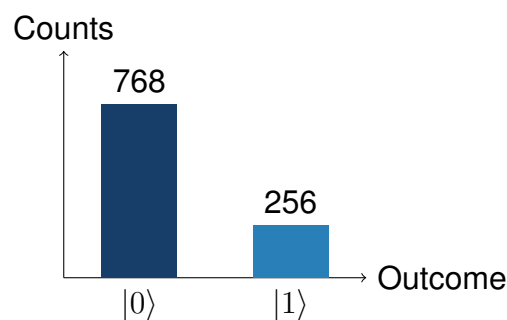
To understand the actual quantum state, we must run the exact same circuit many times and measure the outputs. Each complete execution and measurement is called a **shot**.

Statistical Reconstruction Example

Suppose we prepare a qubit in the state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$.

- Theoretical $P(0) = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4} = 75\%$
- Theoretical $P(1) = \left(\frac{1}{2}\right)^2 = \frac{1}{4} = 25\%$

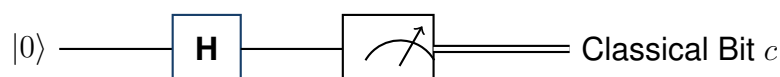
If we run this circuit for **1024 shots** on a quantum computer, the classical output might look like a histogram:



The aggregate statistics from many shots allow us to infer the original probability amplitudes (though phase information is still lost).

4. Circuit Representation and the Z-Basis

In quantum circuit diagrams, measurement is represented by a meter symbol. It takes a quantum wire (solid line) as input and outputs a classical bit wire (often represented as a double line).



The Measurement Basis: By default, almost all quantum hardware measures in the **Computational Basis (Z-basis)**, checking if the qubit is $|0\rangle$ or $|1\rangle$. If you want to measure in a different basis (like the X-basis to check if it's $|+\rangle$ or $|-\rangle$), you must first apply a basis-change gate (like a Hadamard gate) *before* the standard measurement.